

Algebra 1
Unit 9
Lesson 3 Practice

Name Answer key
Date _____
Period _____

Use the information provided in each table to write the equations in vertex form and standard form.

1. Identify the vertex.

(3, -2)

x	1	2	3	4	5
y	18	3	-2	3	18

2. Write the equation for this quadratic in vertex form and standard form.

- a. Vertex form:

$$f(x) = 5(x - 3)^2 - 2$$

- b. Standard form:

$$f(x) = 5x^2 - 30x + 43$$

3. Identify the vertex.

(-1, 3)

4. Write the equation for this quadratic in vertex form and standard form.

- a. Vertex form:

$$f(x) = -(x + 1)^2 + 3$$

- b. Standard form:

$$f(x) = -x^2 - 2x + 2$$

x	y
-4	-6
-3	-1
-2	2
-1	3
0	2

5. Identify the vertex.

(-3, 0)

6. Write the equation for this quadratic.

- a. Vertex form:

$$f(x) = -\frac{1}{2}(x+3)^2 + 0$$

- b. Standard form:

$$f(x) = -\frac{1}{2}x^2 - 3x - 4\frac{1}{2}$$

x	y
-4	$-\frac{1}{2}$
-3	0
-2	$-\frac{1}{2}$
-1	-2
0	$-\frac{9}{2}$
1	-8

7. Identify the vertex.

(1, 2)

x	0	1	2	3	4
y	4	2	4	10	20

8. Write the equation for this quadratic.

- a. Vertex form:

$$f(x) = 2(x-1)^2 + 2$$

- b. Standard form:

$$f(x) = 2x^2 - 4x + 4$$

9. Identify the vertex.

(0, 0)

10. Write the equation for this quadratic.

- a. Vertex form:

$$f(x) = -3(x-0)^2 + 0$$

- b. Standard form:

$$f(x) = -3x^2$$

x	y
-1	-3
0	0
1	-3
2	-12
3	-27